

UNIFYING SPEECH AND LANGUAGE THROUGH CONTINUOUS DUAL-FUSION FOR MEDIUM-RESOURCE LANGUAGES: THE CASES OF HUNGARIAN AND GREEK

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ABSTRACT

In medium-resource languages like Hungarian and Greek, Automatic Speech Recognition (ASR) suffers from scarce paired acoustic data, whereas abundant text data can be leveraged via Large Language Model (LLM) priors. To compensate for this acoustic deficit, we propose a Dual-Fusion architecture that conditions ASR decoding on continuous acoustic information at both the prompt level and inside the LLM’s intermediate layers. Our model bridges a frozen acoustic encoder with pre-trained LLMs via both prompt-level embeddings and deep, continuous-space alignment across multiple intermediate decoder layers. To actively control the semantic-acoustic trade-off, our methodology introduces asymmetric downsampling to preserve fine phonetic boundaries, multi-layer gated cross-attention, attention-weighted layer-wise fusion of the acoustic encoder’s hidden states, and target text perturbation. Evaluated on diverse Hungarian and Greek datasets, our framework reduces Word Error Rate (WER) compared to both standalone acoustic decoders and acoustic-linguistic models like Vox-KriKri, demonstrating consistent performance improvements across both languages.

Index Terms— Automatic Speech Recognition, Speech LLMs, Continuous Fusion, Medium-Resource ASR, Multimodal Alignment

1. INTRODUCTION

Automatic Speech Recognition (ASR) has evolved significantly with the advent of robust end-to-end models like Whisper [1] and highly data-efficient architectures like Canary [2], which achieve state-of-the-art multilingual performance. However, empirical evaluations reveal that these systems frequently exhibit performance degradation when applied to medium-resource and morphologically complex languages, such as Hungarian and Greek. We hypothesize that this drop in accuracy is not solely a failure of acoustic feature extraction; rather, it may partly manifest as orthographic and morphological errors stemming from a lack of deep, language-specific linguistic priors, consistent with prior evidence that external language models can improve low-resource Whisper transcription [3].

To bridge this gap, recent research has shifted toward multimodal Speech-Language Models (SLMs) that couple pre-trained acoustic encoders with Large Language Models (LLMs) [4]. Broadly, this literature has bifurcated into two families of integration strategies [5]: prompt-level approaches that prepend compressed acoustic embeddings as pseudo-text tokens, and cross-attention approaches that let text states attend to acoustic representations at intermediate layers. Early prompt-level strategies sought to minimize architectural complexity: the SLAM-ASR framework [6] demonstrated that a simple, trainable linear projector connecting a frozen encoder like

Whisper to a frozen LLM could yield highly competitive ASR capabilities. SALMONN [7] extended this with a learned Q-Former to compress and bridge audio embeddings into the LLM space. We hypothesize that the rigid temporal pooling used by such prompt-only connectors risks destroying the micro-phonetic boundaries required for highly inflected languages; consistent with this concern, SLAM-ASR has separately been shown to degrade under domain shift and acoustic perturbations [8].

To achieve deeper semantic integration, researchers explored cross-attention fusion. The Whispering LLaMA framework [9] pioneered cross-modal generative error correction by integrating Whisper’s audio embeddings directly into the LLM via cross-attention adapters. A comparative connector study [10] reports that multi-head cross-attention, despite higher parameter cost than linear projectors, can yield better representational alignment than simple connectors under their evaluation. Furthermore, foundational acoustic-language research, such as the Whisper-Flamingo [11] established the efficacy of gated cross-attention to interleave frozen text blocks with continuous multimodal inputs, ensuring stability during generation. In the speech domain, the recent VoxKriKri [12] framework operates fully within a continuous latent space, fusing Whisper’s hidden decoder states with an LLM’s textual representations at a single selected layer, showing that intermediate layer injection yields robust cross-modal alignment for Greek.

A related question is which encoder representations to expose. Rather than passing only an encoder’s final hidden state, SUPERB [13] established that different layers of a self-supervised speech encoder carry different phonetic and semantic information; follow-up work [14] shows that a learned weighted combination over all encoder layers can outperform single-layer extraction. More recent work replaces static weights with input-dependent attention, e.g. Adaptive Layer Attention [15] and TASLA [16]. Finally, a persistent failure mode in fused acoustic-LLM decoders is the “lazy LLM” problem, where the decoder ignores its acoustic information and defaults to its text prior. We adopt a training-time token-substitution technique to mitigate this by deliberately weakening the model’s access to clean linguistic context during teacher forcing; this is conceptually related to, but distinct from, inference-time textual perturbation techniques used to counter hallucination in multimodal LLMs [17].

Building on these findings, we propose a Dual-Fusion architecture that couples a frozen Whisper encoder with language-specialized LLM backbones for Hungarian and Greek. Our methodology combines: (i) a *Dual-Fusion* strategy feeding acoustic information at both the prompt level and via cross-attention, with decoupled downsampling factors; (ii) *multiple injection layers* with per-depth *layer-wise fusion* of all Whisper encoder layers rather

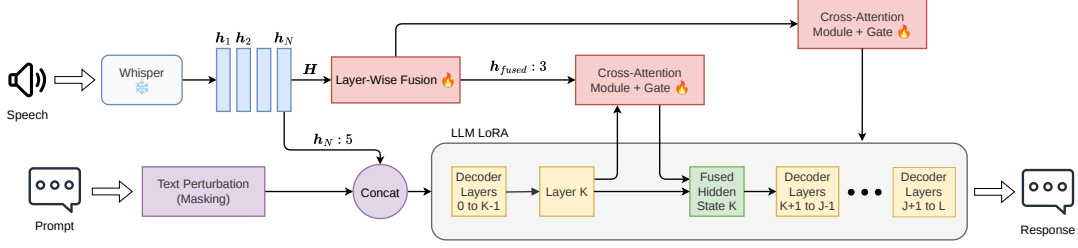


Fig. 1: Dual-fusion architecture: prompt injection, cross-attention, gated fusion, multiple injection layers, and layer-wise fusion.

than only the final state; (iii) *gated cross-attention* scaling injected context with a learnable tanh gate; and (iv) *text perturbation*, substituting random tokens in the teacher-forced target sequence. While each of these ideas has isolated precedent in the work surveyed above, we are not aware of a prior system that combines them within a single continuous, audio-conditioned architecture; we study whether doing so improves performance for both Hungarian and Greek, two typologically distinct medium-resource languages, and report a component-wise ablation (Section 4.5). Evaluated on multiple Hungarian and Greek corpora, and compared against standalone Whisper and against VoxKrikri, our Dual-Fusion models yield consistent Word Error Rate reductions across both languages.

2. METHODOLOGY

The methodology couples a frozen Whisper encoder with a language-specific LLM (Racka-4B for Hungarian, Llama-Krikri-8B-Instruct for Greek) through prompt-level conditioning and intermediate cross-attention. An overview is given in Fig. 1.

2.1. Dual Fusion Strategy

To effectively bridge the temporal mismatch between high-frequency acoustic frames and dense semantic tokens, we deploy a Dual-Fusion Strategy featuring asymmetric downsampling. Acoustic embeddings are integrated at two distinct points: the prompt level and the intermediate cross-attention layers: In the *prompt path*, the final Whisper encoder state h_N is reshaped with a downsampling factor of 5 and passed through a two-layer MLP $\text{Proj}(\cdot)$, then concatenated into the LLM’s input embedding sequence as prefix-like speech tokens:

$$\text{embed}(\text{prefix}) \parallel \text{Proj}(h_N : 5) \parallel \text{embed}(\text{instr}) \parallel \text{embed}(\text{target})$$

In the *cross-attention path*, text states attend to speech-derived keys and values at selected decoder layers. To better preserve short Hungarian suffixes and fast Greek vowel transitions, this path uses a learnable Conv1D downsampler with a smaller factor of 3, retaining a higher temporal resolution than the prompt path for the same underlying encoder states.

2.2. Multiple Injection Layers as Persistent Acoustic Memory

VoxKrikri injects at a single selected depth. We instead inject cross-attention simultaneously after several intermediate decoder blocks (e.g., after blocks [10, 20, 30] of 32 for KriKri, and [15, 25, 35] of 36 for Racka), chosen with a fixed stride of ten blocks so as to sample early-, mid-, and late-decoder representations, since a single early exposure risks being diluted as the hidden state propagates deeper. Repeated injection maintains a persistent acoustic memory,

continually re-grounding generation instead of drifting toward pure language-model continuation.

2.3. Layer-Wise Fusion of Acoustic Representations

VoxKrikri fuses only Whisper’s final hidden state; we instead stack all encoder hidden states $\{h_0, h_1, \dots, h_N\}$, since lower states preserve phonetic detail and higher states carry semantic abstraction. A *LayerWiseAttention* module mean-pools each state’s representation over the time dimension and scores it with a shared MLP, producing an *utterance-level, time-invariant* combination:

$$h_{\text{fused}}^{(j)} = \sum_{l=0}^N \alpha_l^{(j)} h_l, \quad \alpha_l^{(j)} = \text{softmax}_l \left(v^\top \tanh(W \bar{h}_l) \right)$$

2.4. Gated Cross-Attention

Injecting acoustic context at multiple layers along two pathways increases the risk that acoustic signal overwhelms the LLM’s own generative flow, especially for tokens that are purely syntactic (punctuation, function words) and should be governed by linguistic, not acoustic, evidence. Following the zero-initialized gating principle introduced by Flamingo [18], each injection layer scales its cross-attention output with a single learnable scalar gate $g^{(j)}$:

$$h_{\text{fused}} = h_{\text{LLM}} + \tanh \left(g^{(j)} \right) C_{\text{audio}}$$

2.5. Text Perturbation

This mechanism counters the “lazy LM” problem; where teacher-forced text alone lets the decoder ignore the acoustic information. During teacher forcing, each eligible target-history token is replaced by the tokenizer pad ID with probability $p = 10\%$ while the clean next-token labels are retained. This weakens textual history available for subsequent predictions and encourages greater use of acoustic conditioning.

3. EXPERIMENTAL SETUP

3.1. Backbones and Injection Configuration

Both language configurations adopt the frozen Whisper-large-v3 [1] encoder (636M parameters). For Greek, the LLM backbone is Llama-KriKri-8B-Instruct [19] (8B); for Hungarian, elte-nlp/Racka-4B [20] (4B). The trainable bridge: reshape adapter (19M), a Conv1D downsampler (5M) shared across all injection layers (Section 2.2), cross-attention modules (59M), and layer-fusion module (1M) — totals 84M parameters (0.9% and 1.75% of the full model in Greek and Hungarian respectively), making Stage 1 substantially

cheaper than full fine-tuning. Stage 2 additionally trains 133M LoRA parameters on the Racka-4B LLM and ?M on the KriKri LLM. Each cross-attention module uses 32 attention heads.

3.2. 2-Stage LoRA Curriculum and Training Hyperparameters

In Stage 1, the LLM and Whisper encoder are fully frozen and only the bridge network is optimized, for 45000 steps, with learning rate 10^{-4} , an effective batch size of 64 (per-device batch size 4 with 4 gradient-accumulation steps across 4 GPUs), bf16 mixed precision, and the AdamW [21] optimizer under a cosine scheduler with warmup. In Stage 2, LoRA is activated on the LLM with rank $r = 64$, scaling $\alpha = 2r = 128$, dropout 0.1, targeting the attention projections and the gated-MLP up/down/gate projections [22]; the bridge network trained in Stage 1 (adapter, downsampler, and cross-attention modules) is frozen for the remainder of training so that LoRA adapts the LLM to the now-fixed acoustic interface rather than co-adapting with a still-moving bridge. This stage runs for 45000 steps, too, at learning rate 10^{-4} . The LLM is fine-tuned under 4-bit NF4 quantization with double quantization to reduce memory footprint.

3.3. Corpora

Tables 1 and 2 list the Greek (Common Voice [23], FLEURS [24], HParl [25], TEDx [26], Logotypographia [27]) and Hungarian (Common Voice [23], FLEURS [24], Speech Massive [28, 29], VoxPopuli [30], Yodas [31], and two commercially licensed corpora, anonymized here as Internal-1 and Internal-2) corpora, spanning formal, read, and spontaneous registers. Each corpus is split into train/validation/test following a fixed priority: (1) the creators’ own official splits, if provided; (2) otherwise, a leave-subject-out (LSO) split by speaker/subject ID, if the corpus is subject-segmented, ensuring no speaker overlaps train and test, using a 60/20/20 ratio; (3) otherwise, a random split using a 60/20/20 ratio, seeded for reproducibility. Since dataset sizes vary by orders of magnitude, training batches are constructed via interleaved sampling across corpora with a temperature factor of 1.2, downweighting the largest datasets.

3.4. Preprocessing and Prompting

Transcripts are normalized before training/scoring: removal of speaker-tag prefixes and transcriber annotations, and verbalization of digits into their spoken word forms. Each utterance is additionally conditioned on a natural-language epistemic prompt telling the model what transcription style to produce — [PROMPT_CLEAN] (accurate, clean transcript), [PROMPT_VERBATIM] (word-for-word transcript, disfluencies retained), or [PROMPT_FORMAL] (strictly formal/official transcript): each corpus is deterministically assigned one style matching its own reference-transcription convention (falling back to [PROMPT_CLEAN] if unspecified), applied consistently at training and inference. During training, Whisper’s input features are also regularized with SpecAugment [32], which we found helps prevent the bridge network from overfitting.

4. RESULTS AND DISCUSSION

4.1. Greek: Comparison Against Baselines

Table 1 compares against full-context VoxKrikri [12] and Whisper-v3 across five Greek corpora. Our model outperforms VoxKrikri on Common Voice ($\Delta = -4.74$), HParl ($\Delta = -4.81$), and Logotypographia ($\Delta = -2.80$), but FLEURS is the one corpus

where VoxKrikri’s simpler single-stage, single-injection design wins: on FLEURS, VoxKrikri (full) (9.33) outperforms ours (10.9, $\Delta = +1.57$), though we still improve over Whisper-v3 (11.02) on this dataset. We attribute this to FLEURS’s high density of translated foreign named entities, which our LLM’s strong linguistic prior tends to mis-transliterate into valid but incorrect Greek vocabulary, consistent with the failure mode discussed in Section 4.4.

Table 1: Greek WER (%) comparison against baselines.

Dataset	Ours	Δ	Vox (full)	Whisper
Common Voice(34h) [23]	7.3	-4.74	12.04	13.98
FLEURS(12h) [24]	10.9	+1.57	9.33	11.02
HParl(120h) [25]	8.2	-4.81	13.01	16.99
Logotypographia(52h) [27]	6.6	-2.80	9.4	10.86
TEDx(30h) [26]	17.8	—	—	—

4.2. Hungarian: Comparison Against Baselines

Table 2 reports and compares the WER scores across all five public Hungarian corpora against the original multilingual Whisper large-v3 [1]. We deliberately exclude the Hungarian-specific Whisper-Hu [33] checkpoint, whose model documentation discloses fine-tuning on the training, validation, and test splits of VoxPopuli, Common Voice, and Speech-Massive, making comparison on those benchmarks non-zero-shot and unfair; Our model outperforms Whisper on four of the five comparable corpora (Table 2), with the largest margins on FLEURS (10.2 vs. 37.8, $\Delta = -27.6$) and Speech Massive (38.2 vs. 59.1, $\Delta = -20.9$), and further wins on Common Voice ($\Delta = -6.9$), VoxPopuli ($\Delta = -8.0$). The sole loss is Yodas, where plain Whisper remains stronger (11.5 vs. 8.1, $\Delta = +3.4$). This loss likely reflects reference provenance rather than a real shortfall: Yodas transcripts are pseudo-labeled by faster-whisper-large-v3 itself, so comparing us to zero-shot Whisper on Whisper-generated references is close to self-referential evaluation.

Table 2: Hungarian WER (%) comparison against the original Whisper baseline.

Dataset	Ours	Δ	Whisper
Common Voice(180h) [23]	9.0	-6.9	15.9
FLEURS(12h) [24]	10.2	-27.6	37.8
Speech Massive(120h) [28]	38.2	-20.9	59.1
VoxPopuli(120h) [30]	9.8	-8.0	17.8
Yodas(150h) [31]	11.5	+3.4	8.1

4.3. Model Interpretability

To explore *why* injecting cross-attention at three depths (Section 2.2) may behave, Fig. 2 visualizes raw cross-attention weights of the three Racka injection layers ([15, 25, 35]) as word-level alignment maps (summed subword attention per word, averaged over heads) for a single representative Hungarian test utterance: (i) The earliest layer (15) shows attention concentrated in a few narrow, high-intensity spikes rather than a monotonic trace, consistent with picking out locally salient acoustic events (ii) The middle layer (25) shows a near-monotonic diagonal, consistent with fine-grained temporal grounding; notably, this layer sits at $\sim 69\%$ relative depth of

Racka’s 36 decoder layers, close to the 21st of 32 layers independently reported as the strongest single injection point for VoxKrikri among $\{1, 9, 15, 21, 30\}$ [12] (iii) The latest layer (35) shows diffuse, non-monotonic attention with several positions attended from many temporally distant words This depth specialization (coarse detection, precise grounding, global re-grounding) supports the persistent-acoustic-memory motivation for multi-layer injection (Section 2.2). We report this as a qualitative observation, not a quantitative claim, which would require aggregate diagonality or alignment-error metrics over a representative sample; we leave this quantification to future work.

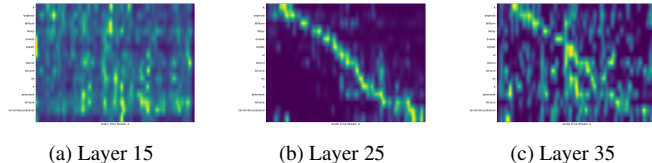


Fig. 2: Word-level cross-attention alignment for a representative Hungarian test utterance, shown separately for each of the three injection layers

4.4. Limitations

Our qualitative analysis surfaces a recurring failure mode: because the fused LLM carries strong linguistic priors, it sometimes acts less like a verbatim transcriber and more like a “copy editor”. Observed behaviors include silently dropping hesitation markers and dysfluencies, normalizing colloquial contractions to their formal written form, occasionally completing thoughts on abruptly cut-off audio, and mis-transliterating foreign proper names, inflating WER on disfluency-rich, verbatim-style corpora even when the output is fluent and largely faithful in meaning. Our evaluation is also limited to two medium-resource languages, a single frozen acoustic encoder (Whisper-large-v3); generalization to other languages, encoders, or truly low-resource settings remains untested.

4.5. Ablation Study

Table 3 outlines the progressive ablation needed to attribute the reported gains to individual design choices rather than their combination

Table 3: Ablation for Hungarian, validation-set WER (%)

Configuration	WER (%)	Δ vs. Full
Full model	21.4	–
With 2 injection layers (15, 25)	<i>TBD</i>	<i>TBD</i>
With 2 injection layers (25, 35)	<i>TBD</i>	<i>TBD</i>
With 1 injection layer (25)	<i>TBD</i>	<i>TBD</i>
Without layer-wise acoustic fusion	23.2	<i>TBD</i>
With input-independent layer weights	22.5	<i>TBD</i>
Without gated cross-attention	<i>TBD</i>	<i>TBD</i>
Without text perturbation, $p = 0\%$	22.5	<i>TBD</i>
With text perturbation, $p = 20\%$	22.1	<i>TBD</i>
With text perturbation, $p = 30\%$	22.5	<i>TBD</i>

5. CONCLUSION & FUTURE WORK

We presented a Continuous Dual-Fusion architecture coupling a frozen Whisper encoder with language-specialized LLMs for medium-resource ASR in Hungarian and Greek, using multi-layer gated cross-attention, layer-wise acoustic pooling, and text perturbation to ground generation in the acoustic signal without discarding linguistic priors. Across nine Hungarian and five Greek corpora, our model reduces WER over standalone Whisper on all but one Hungarian corpus (Table 2) and beats the strongest published baseline, VoxKrikri, on three of four comparable Greek corpora (Table 1).

Beyond adding more languages and encoders, we plan to: (i) incorporate acoustic data from higher-resource languages (e.g., English, German for Hungarian and English, Cypriot for Greek) to strengthen shared representations while confining foreign-language exposure to the encoder/cross-attention pathway and monitoring for code-switching leakage; (ii) use TTS-synthesized utterances to expose the linguistic side to target-language words absent from our corpora, particularly for morphologically rich Hungarian; and (iii) evaluate an audio-forecasting head, already implemented but not yet assessed, that predicts the next downsampled acoustic embedding from the LLM’s hidden states at injected audio-token positions, which could also serve as transcript-free self-supervised pretraining over unlabeled speech, , directly addressing the paired-data scarcity motivating this work.

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